

Exhaustive inferences and additive presuppositions: The interplay of focus operators and contrastive intonation

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Abstract

The research reported here investigates the interplay of the overt focus operators *only* and *also* with contrastive intonation. Our experiment consisted of two parts: a naturalness rating and a delayed truth value judgement task. The results of the rating task indicate that focus particles are perceived as equally natural combined with an H* accent and an L+H* accent. The truth value judgement task showed that L+H* accents reinforce exhaustive inferences to the same extent as the overt operator *only*. In the case of *also*, listeners infer an additive presupposition even if such a presupposition is not explicitly introduced in the context. Overall, the results indicate that once an overt focus operator is present in an utterance, L+H* accents do not contribute any further information. In utterances without overt focus operators, on the other hand, L+H* accents act like an *only* operator.

Keywords: Focus operators; contrastive intonation; exhaustive inferences; presuppositions.

Introduction

Successful communication involves drawing inferences about implicitly conveyed content of a message. For example, sentence (1) is usually interpreted exhaustively such that the speaker liked Tübingen but did not like Konstanz, even though this information is not explicitly stated. In cases such as (1), a so-called exhaustive implicature arises through pragmatic enrichment of meaning (see for example Groenendijk & Stokhof, 1984; Van Rooij & Schulz, 2004).

- (1) Context: Last year, I went to Tübingen and Konstanz.
I liked Tübingen.

What is implicitly conveyed by the utterance above is part of the lexical semantics of *only*, that is, its conventional meaning. For example, a sentence like 'I only liked Tübingen' asserts that the speaker did not like Konstanz (e.g., König, 1991). With overt *only*, the exhaustive inference always arises and it cannot be cancelled. In the case of an exhaustive implicature, on the other hand, the speaker could continue the discourse by stating 'In fact, I liked Konstanz too' cancelling the exhaustive inference. Nevertheless, we could think of the utterance in (1) as containing a silent *only* operator, accounting for the fact that a similar exhaustive meaning arises as with overt *only* (Groenendijk & Stokhof, 1984; Van Rooij & Schulz, 2004; Chierchia, 2004).

Such exhaustive inferences are reinforced by contrastive intonation (e.g., Chevallier et al., 2008, see also the next section). According to Pierrehumbert and Hirschberg (1990),

certain intonational contours are associated with a contrastive interpretation.¹ Pierrehumbert and Hirschberg (1990) distinguish between a complex L+H* accent consisting of a low initial target followed by steep rise and a simplex H* accent with a single high pitch accent. The authors assume that L+H* accents trigger a contrastive inference while H* accents have a non-contrastive meaning.²

In the present study, we compare the exhaustive inferences triggered by contrastive intonation, non-contrastive intonation, the overt focus operator *only* and a combination of *only* with either type of intonation. One possible hypothesis is that particles like *only* require a contrastive accent (see for example Jacobs, 1983; Dimitrova, 2012). Alternatively, contrastive intonation might not add any further information once an overt operator is present in an utterance. In other words, contrastive intonation might act like a covert *only* operator (Groenendijk & Stokhof, 1984; Van Rooij & Schulz, 2004; Chierchia, 2004).

We further compare exhaustive inferences to the additive presupposition associated with *also*. Particles like *also* presuppose that an alternative holds and add the assertion that the proposition is true for the focused element (König, 1991). For example, the sentence 'I also liked Konstanz' presupposes that the speaker liked Tübingen in addition to Konstanz. *also* is considered a 'hard' presupposition trigger in that it always triggers a presupposition, which in most contexts cannot be easily accommodated (Abusch, 2002). Since *also* is an additive particle, contrastive intonation should not affect speaker's inferences. Another possibility could be that *also* is incompatible with a contrastive accent. In that case, utterances that combine *also* with an L+H* accent might be judged as less felicitous than utterances with *also* and H* accent.

Previous research

Focus particles and contrastive intonation In order to determine how focus particles interact with contrastive intonation, it is important to find out with what accent type focus

¹We use the terms contrastivity and exhaustivity synonymously here without intending any theoretical implications. In our examples contrastive and exhaustive inferences coincide (e.g., A and not B is equivalent to A and nothing else).

²Note, however, that there is an open theoretical debate whether those two accent types represent distinct categories or are variants of the same accent (see for example Krahmer & Swerts, 2001)

particles most naturally combine with.³ Two previous studies provide some insights into this question. Sudhoff (2010) conducted a production and a perception experiment to investigate whether focus particles induce contrastive focus or rather interact with a given focus background structure. The results showed that focus particles were produced with both variants, either contrastive or non-contrastive pitch accents (independent of whether *only*, *even* or *also* was used). In the perception study, participants had to indicate whether a contrast was present or not. The results revealed that the contrast judgements were independent of whether and which specific focus particle was used. Therefore, Sudhoff (2010) suggested that the phonetic realization of a focused element is dependent on the focus background structure of the linguistic context and not on the presence or absence of a focus particle.

Dimitrova (2012) used the ERP technique to investigate how sentences with the dutch particle '*alleen*' ('only') are processed in contexts with and without contrastive accenting. She found a positive ERP component around 200-500 ms for contrastive accents compared to utterances without an accent. The particle *only* modulated the processing of accented constituents (compared with sentences without *only*): the accent positivity was delayed in sentences with *only* and additional early anterior negativities and late left anterior positivities were triggered. According to Dimitrova, the results suggest that sentences with *only* trigger additional processing costs, reflecting the expectation of an accent or the fact that *only* may require a contrastive accent. To summarize, it is currently unclear whether focus particles require a contrastive accent or are equally felicitous with contrastive and non-contrastive accents.

Exhaustive inferences and additive presuppositions Regarding the associated meaning of focus particles and contrastive accents, a study by Onea and Beaver (2011) showed that participants draw more exhaustivity inferences in sentences with *only* compared to simple intonational focus in German (but no exact information about the accent type is given). Contrastive intonation seems to reinforce such exhaustive inferences. For example, a study by Chevallier et al. (2008) showed that participants are more likely to interpret disjunctive *or* exclusively when this word receives contrastive stress.

Fraundorf, Watson, and Benjamin (2010) provided evidence that L+H* accents benefit long-term representations of exhaustive inferences in sentences without any lexical implicature triggers. They compared H* and L+H* pitch accents in a delayed recognition memory task involving a truth value judgement. The results indicated that the L+H* accent increased the number of correct rejections of statements about a mentioned alternative. This indicates that the L+H* accent triggered an exhaustive inference, which participants remembered even a day later.

There is also a growing body of research investigating

the processing of exhaustive inferences, additive presuppositions and the role of contrastive intonation in online processing. Tomlinson, Bailey, and Bott (2013) showed that contrastive accents facilitate the speed with which listeners arrive at an exhaustive interpretation. A recent visual world study by Schwarz (in press) compared the exhaustive assertion triggered by *only* to the additive presupposition of *also*. The eyegaze patterns indicated that listeners evaluated presuppositions more immediately than asserted content. So, the additive presupposition of *also* might be integrated earlier into sentence meaning than is the exhaustive assertion associated with *only* (see also (Kim, 2012) for work comparing *only* and *also* in online processing). Schwarz (in press) assumes this result to be most consistent with semantic theories that take presuppositions as a 'pre-condition' on the common ground. In summary, processing results may provide important insights into theories of semantic and pragmatic meaning. However, we focus on the end result of processing, that is a speaker's final interpretation here.

The present experiment

In the experiment presented here, we explore the interplay of overt focus operators with contrastive intonation, comparing the inferences triggered by (i) L+H* accents, (ii) H* accents, (iii) the exclusive particle *only*, (iv) the additive particle *also*, and (v) a combination of focus operators with either type of intonation.

We pursue two research questions. First, we address the question what accent types focus particles most naturally combine with by using acceptability ratings. Second, we are interested in whether listeners draw the same amount of exhaustive inferences from the truth-functional operator *only* compared to a contrastive accent and whether a combination of the two leads to additive effects. In the case of *also*, contrastive intonation should not affect judgements.

Participants in our experiment were presented with auditory discourses in German. In a context sentence, a contrast set with two persons was introduced (e.g., 'The judge and the witness followed the argument'). In a critical sentence, one of those persons was mentioned again carrying either an H* accent or an L+H* accent without any preceding particle (e.g., 'The judge believed the defendant'). Four further conditions were created in which the critical sentences started with the particle *only* or *also* combined with either the H* or L+H* accent on the following critical noun (e.g., 'Only the judge believed the defendant').

After participants were presented with the stimuli, they performed a naturalness rating. Further, we assessed participants' memory for inferences about the alternative mentioned in the first context sentence (i.e. not the critical noun itself). After a delay of about 1.5 minutes (5 trials) participants were presented with statements about the discourses and had to judge whether these statements were true or false regarding the content of a given story.⁴

³We thank Duane Watson for a helpful discussion concerning this point.

⁴We use a delayed truth value judgement task in order to avoid

If participants interpret the discourses exhaustively, they are expected to indicate that the statement about the mentioned alternative is false (especially in the conditions with *only* and L+H* accent). In the case of the particle *also*, we expect participants to infer that the alternative statement is true. Our stimuli do not introduce the presupposition of *also* explicitly. Nevertheless, we expect listeners to infer this presupposition.

Methods

Participants A total of 24 native speakers of German (18 female and 6 male, mean age 25.25 years, age range 20-31) were recruited from a participant pool at the Institute of Psychology of Humboldt University. In total, 25 participants had participated in the study, but one of the subjects was bilingual (English as first native language) and was therefore excluded from further analyses. Participants were paid seven Euros in compensation. None of them reported any vision or hearing difficulties.

Materials The discourses followed the structure presented in (2). A first context sentence introduced two persons. A second critical sentence mentioned one of the two persons again describing an action. A third sentence continued the narrative. Note that the discourse did not introduce the presupposition associated with *also*. For example, there was no sentence stating that somebody else (the witness) believed the defendant.

- (2) *Context sentence:*
Der Richter und der Zeuge verfolgten die Beweisführung.
'The judge and witness followed the argument.'
Critical sentence:
(Nur)/(Auch) der Richter/RICHTER glaubte dem Angeklagten.
'(Only)/(Also)⁵ the judge/JUDGE believed the defendant.'
Continuation sentence:
Er verkündete das Urteil.
'He announced the verdict.'

The experimental design fully-crossed the factors particle condition and accent type resulting in the six conditions summarized in Table 1.

All sentences were recorded in a sound-proof room by the first author of this paper. The critical sentences were spoken in four different versions, with an H* accent, an L+H* accent, the particle *only* and *also*. Figure 1 shows the averaged intonational contour of the accented syllable of the critical

ceiling effects and to eliminate possible differences in processing across conditions. That is, we are interested in the final interpretation of the utterances regardless of the path/process with which listeners arrive at a particular interpretation. See Schwarz (in press) and Kim (2012) for insights into the processing of additive and exclusive focus particles.

⁵Unlike in English, in German *also* can occur in this syntactic position.

Table 1: Overview of conditions

Accent/Particle	<i>only</i>	<i>also</i>	<i>bare</i>
H*	only-H*	also-H*	H*
L+H*	only-L+H*	also-L+H*	L+H*

noun across accent type conditions. Acoustic measurements were conducted revealing that the condition with L+H* accent had a longer duration, a greater intensity and a higher pitch excursion. Details of the acoustic analysis are available from Gotzner, Spalek, and Wartenburger (2013). To create all six versions of an experimental item, we cross-spliced the particles from the originally recorded stimuli into either the sentence with H* accent or L+H*.

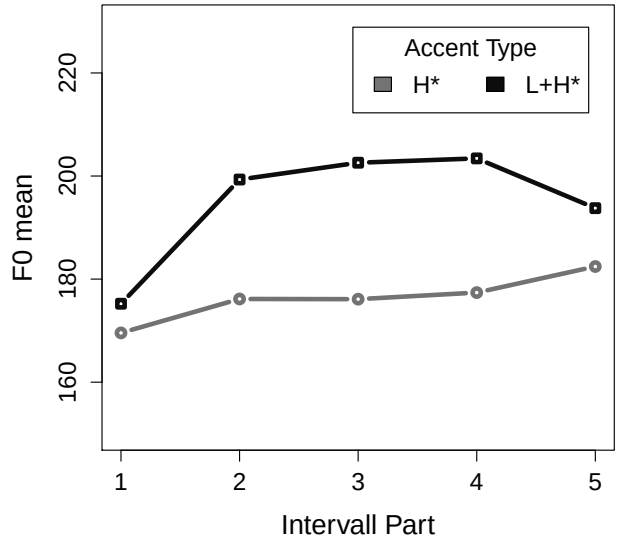


Figure 1: Accented syllable of critical noun (e.g., judge) across H* and L+H* conditions

In total, there were 60 critical items corresponding to the structure presented in (2). Participants were probed with critical statements concerning the experimental items. For example, participants were asked to judge whether the statement "The witness believed the defendant" was true or false. The critical statements were always about the alternative mentioned in the context sentence. For example, if the critical sentence was "The judge believed the defendant" the statement mentioned the witness.

We further added 20 filler items with the same structure as the experimental items and a set of 40 filler items with mild pragmatic violations so that participants rated acceptable and less acceptable items. An example of an unacceptable filler item is shown in (3). Participants were probed with general statements about the narrative in the filler items (not involv-

ing alternative statements). These general statements could concern any of the three sentences and any of the characters.

- (3) Marie und Josi waren im Pferdestall.
'Marie and Josi were in the horse barn.'
Marie wollte die Pferde striegeln.
'Marie wanted to groom the horses.'
Sie bereitete immer das Essen vor.
'She always prepared the food.'

Six experimental lists were created, rotating through the accent and particle conditions. To each list, the 60 filler items were added totalling in 120 items per list. Each participant received a different randomization with at most two filler items appearing in a row and the different experimental conditions being repeated at most twice.

Procedure An on-screen instruction explained the structure of the experiment. The instructions told the participants that they will have to rate the naturalness of auditory stimuli and to remember the content of the stories. They were told to judge how coherent and natural the stories were on a scale from 1 (not at all acceptable) to 7 (very acceptable). We further asked them to indicate whether a statement was true or false regarding the content of a particular story. We told participants that the statements could be implicit in the story. Five practice trials were administered before the experiment started.

The experiment consisted of two paired phases: (a) rating phases and (b) truth value judgement phases. Figure 2 represents the structure of one experimental block. Rating and judgement phases interchanged every five items. After a total of five rating trials, a screen informed participants that the judgement phase would start and they had to judge the given statements as true or false.

Each experimental trial began with a central fixation cross displayed for 500 ms. Then, participants heard an item over headphones. Subsequently, they saw another fixation cross for 500 ms immediately followed by a scale from 1 to 7. Participants had to use a left and right button to browse through the array of numbers. By pressing a third button, they confirmed the selected number. Participants had a time window of up to 8000 ms for the rating. One presentation block lasted about 1.5 minutes.

In the judgement phase, a fixation cross was displayed for 500 ms and then a blank screen was shown for 100 ms, immediately followed by the statement. A time window of up to 10000 ms was allowed for the judgement. The statement was coloured green so that participants could easily identify the judgement phase. The sequence of statements corresponded to the order of presentation of the auditory stimuli. After 5 judgement trials, a screen announced the start of the next presentation/rating block.

In total, there were 24 short blocks. The entire experiment lasted about 45 minutes, including the post experimental questionnaire. In the questionnaire, none of the participants seemed to have been aware of the purpose of the experiment.

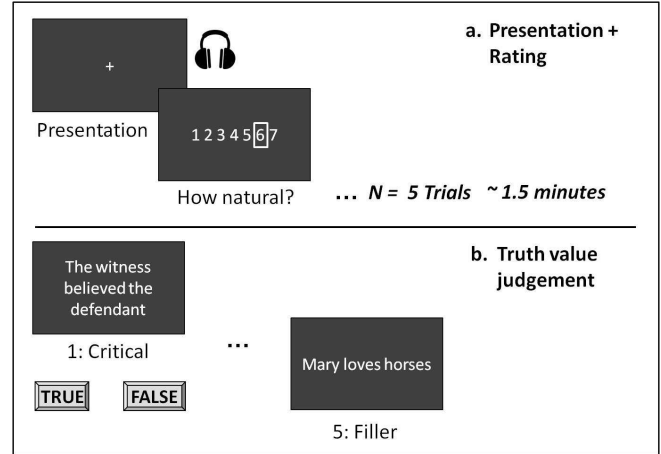


Figure 2: Trial sequence of one experimental block

Results

Naturalness rating

Table 2 shows the mean ratings across conditions. The average rating of the unacceptable filler items was 2.5.

Table 2: Mean ratings across conditions

Accent/Particle	<i>only</i>	<i>also</i>	<i>bare</i>
H*	5.6	5.7	5.8
L+H*	5.9	5.7	5.9

We computed a mixed model with the rating scores including the factors particle condition, accent type, an interaction of these two factors and random effects for subjects and items. The bare H* condition served as reference level. The model did not reveal any significant differences across conditions (all p's > .3).

The results of the mixed model are summarized in Table 3. The lack of significant differences indicates that the combination of the different particles with either an H* or an L+H* accent was perceived as equally felicitous. Further, note that the discourse with *also* was felicitous even though the additive presupposition was not explicitly introduced.

Table 3: Summary of mixed effects model for naturalness ratings (n = 1920, log-likelihood = -3256)

	β	SE	t	p
(Intercept)	5.84936	0.2032	28.783	0.0001
L+H*	0.06731	0.22169	0.304	0.7468
only-L+H*	-0.20353	0.2177	-0.935	0.3072
also-L+H*	-0.10234	0.2177	-0.47	0.6016
only-L+H*:H*	0.18429	0.3107	0.593	0.5132
also-L+H*:H*	-0.08997	0.3107	-0.29	0.7556

Truth value judgement

The mean percentages of TRUE answers from the judgement task are found in Table 4. Figure 3 shows an interaction plot of the results.

Table 4: Mean percent of TRUE responses across conditions

Particle/Accent	only	also	bare
H*	18.3 %	74.6 %	35.4 %
L+H*	13.7 %	74.2 %	21.3 %

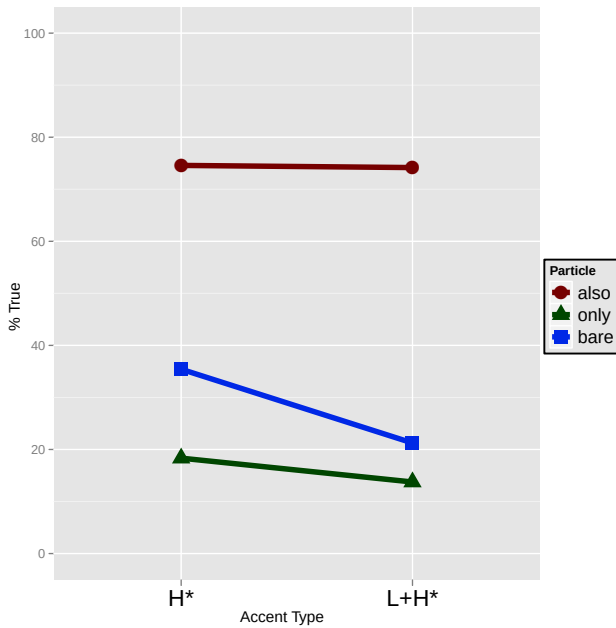


Figure 3: Mean percent of TRUE answers across conditions (circle = *also*, square = *bare*, triangle = *only*).

The responses were analyzed with logit mixed effects models using the bare H* condition as reference level. The results showed that participants gave significantly more FALSE responses in the condition with L+H* than H* accent ($\beta = -.80$, $SE = .30$, $z = -2.64$, $p < .01$) as well as comparing *only* and the H* condition ($\beta = -1.00$, $SE = .30$, $z = -3.25$, $p < .01$). There were significantly more TRUE responses in the condition with *also* than with H* accent ($\beta = 1.89$, $SE = .30$, $z = 6.33$, $p < .0001$). There was also a marginal interaction of *also* with accent type ($\beta = 0.80$, $SE = .43$, $z = 1.85$, $p = .065$). The results of the mixed model are detailed in Table 5.

A post hoc Tukey test revealed that there was no effect of accent type in the *only* and *also* particle conditions. That is, responses did not differ in the H*-only condition and L+H*-only condition ($p = .9$). Similarly, there was no difference between the H+ *also* and L+H*-*also* condition ($p = .99$). Moreover, neither the difference between only-L+H* and the bare

Table 5: Summary of mixed effects model for judgement task ($n = 1440$, log-likelihood = -731.9)

	β	SE	z	p
(Intercept)	-0.6819	0.2453	-2.78	0.00543
L+H*	-0.8041	0.3044	-2.641	0.00826
only-L+H*	-1.0074	0.3098	-3.252	0.00115
also-L+H*	1.8938	0.2994	6.325	0.0001
only-L+H*:H*	0.4281	0.4553	0.94	0.34704
also-L+H*:H*	0.7961	0.4311	1.847	0.06478

L+H* accent nor between only-H* and the bare L+H* was significant ($p = .98$ and $p = .5$, respectively).

We further evaluated whether the bare H* condition differed from chance level. A t test showed that participants gave reliably more FALSE responses than expected by chance ($t = -3.39$, $p < 0.01$).

Discussion

The rating data presented here indicate that the use of a particular focus particle was perceived as equally natural with an H* pitch accent and an L+H* accent on the focused element (and the different bare accent types were also rated the same). This finding is in line with the study by Sudhoff (2010) and it suggests that focus particles do not necessarily require a contrastive pitch accent.

An objection might be that the lack of acceptability differences indicates that participants did not perceive the difference between our intended accent conditions. Such an objection is ruled out by data from the truth value judgement task: Participants computed more exhaustive inferences in the condition with L+H* compared to H*, indicating that the L+H* accent was perceived as more contrastive than the H* accent.

The judgement data replicate the finding by Fraundorf et al. (2010) that participants compute and encode more exhaustive inferences when exposed to contrastive accents compared to non-contrastive accents. Interestingly, participants were equally able to derive such an inference in the L+H* condition and the explicit *only* condition and the effects of accenting and focus particles were not additive (i.e., only-L+H* did not differ from only-H*). This finding indicates that the L+H* has the same function as the overt *only* operator. We are not assuming that L+H* accents and the explicit *only* operators do not differ on any level. However, it seems that the exhaustive inference triggered is equally strong in both cases. Therefore, one might argue that the L+H* accent acts as an *only* operator. This finding is line with theories assuming that implicatures are triggered by a silent *only/exh* operator (Groenendijk & Stokhof, 1984; Van Rooij & Schulz, 2004; Chierchia, 2004).⁶

⁶Note, however, that we are not claiming that the data are incompatible with the Gricean view, since we did not pit theories of implicature against each other.

In this respect, it is interesting to note that participants gave more FALSE responses in the H* condition than is expected by chance level. This suggests that a similar process might have been triggered with the H* accent and that this process was reinforced by the L+H* accent. Even without the L+H* accent, one could infer that the stronger statement does not hold (e.g., The judge and the witness believed the defendant). The L+H* accent increases the likelihood of the exhaustive implicature to occur and possibly the speed with which listeners arrive at the inference (see for example Tomlinson et al., 2013). Regarding the function of pitch accents, such a result might be more compatible with a theory that views the L+H* accent as the more contrastive variant of the H* accent (see for example Krahmer & Swerts, 2001) rather than the categorical view proposed by Pierrehumbert and Hirschberg (1990).

Our study further revealed that listeners infer the presupposition of *also* even if it is not contextually introduced. *Also* is considered as a hard presupposition trigger (e.g., Abusch, 2002). This means that (i) the presupposition of *also* is always triggered and (ii) that it is not easily accommodated (as compared with for example the presupposition of the definite article). The structures we used did not introduce the presupposition of *also*. Nevertheless, participants were willing to infer this presupposition. This is evident in the high amount of TRUE responses in the condition with *also* as well as the high naturalness ratings. In fact, ratings were as high as for the most neutral condition with the H* accent.

This finding indicates that, at least in certain contexts, presupposition accommodation of hard triggers might be easier than previously thought or at least not impossible. We propose that this result is due to the structural position in which *also* occurred in our sentences. Krifka (1993) notes that in sentence initial position (contrastive topic position), it is possible to accommodate an implicit question like 'What happened?'. We assume that since *also* was not embedded in the critical sentence and because a possible antecedent was close in the context sentence, presupposition accommodation was possible. We predict that if one tested a sentence with *also* in object position (e.g., 'Der Richter glaubte auch dem Angeklagten') in the same experimental paradigm acceptability and the amount of TRUE responses would decrease profoundly. In future research, we aim to examine the impact of information structure and the structural position of *also* on presupposition accommodation.

Conclusions

The study reported here provides insights into the impact of information structure, in particular contrastive intonation, on exhaustive inferences. First, it indicates that intonation has an important impact on inferencing (in conjunction with other recent studies). Second, the study indicated that L+H* accents act like implicit *only* operators and were found to be equally effective as overt *only* operators. If an overt focus operator was present, contrastive accents, however, did not seem

to contribute any further information. A third interesting finding is that, in certain structural positions, the presupposition of *also* can be accommodated.

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